

A review on image-based approaches for breast cancer detection, segmentation, and classification

Zahra Rezaei¹

Department of Computer Science, Marvdasht Branch, Islamic Azad University, Marvdasht, Iran

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ABSTRACT

The breast cancer as the most life-threatening disease among the woman has emerged in the worldwide. It is supposed that the early testing and treatment for breast cancer detection would be avoided the surgeries and increase the survival rate. A variety of research studies have motivated to improve the diagnostic methods for early diagnosis of breast cancer. This study investigates the automatic and semi-automatic image-based approaches for breast cancer diagnosis. The scope of this research has limited to the images based diagnosis application journal that are published between 2016 and 2020 years. The principles and associated risk factors for diagnosis the breast cancer and existing imaging techniques are presented. The steps of diagnosis including preprocessing, segmentation, extracting tumor features, and tumor classification are investigated. The publicly available datasets for breast imaging are briefly introduced as well. The application issues, challenges of breast imaging technologies and future directions are discussed. Based on the detailed study, most proposed methods use one type of imaging modalities, however, the doctor need to investigate the multiple imaging techniques to accurate diagnosis and effective treatment. Moreover, handling the multiple imaging require the processing of big data using a cluster computing framework.

1. Introduction

The cancer defines conditions leads to the tumor development (Kandlikar et al., 2017).

In order to prevent, diagnosis, and treatment of breast cancer, different imaging modalities have been developed. The computer-based imaging technologies have important role to help the pathologists for decision making (Puri, Lloyd, & Bui, 2016). Previously, experts manually examined the images to investigate the Microcalcifications (MCs) and malignant tissues. Currently, the advanced algorithms for diagnosis purpose have been developed (Saad, Khadour, & Kanafani, 2016). The intelligent Computer-Aided Diagnosis (CAD) system are developed to assist the radiologists for diagnosis of the abnormalities (Shi, Zhong, Rampun, & Wang, 2018). Automatic segmentation of breast tumor is a vital step. In order to improve the segmentation performance, prior information can be included. However, because of complex characteristics of tumor, prior knowledge, including intensity, texture, and shape may be inadequate (Xi et al., 2017). The medical image segmentation methods is still open problem due to the low accuracy, high complexity, low robustness, and lack of versatility (Yang, 2017). To detect the

cancerous regions, each pixel of breast tissue can be categorized as the malignant or benign. However, pixel classification is quite time consuming due to the processing massive dataset (Xi et al., 2017). Extracting the useful patterns from a huge dataset would be difficult, therefore data sampling techniques is developed to show the data behavior in small portion of data (Salahuddin et al., 2018). These techniques reduce the error sampling and preserve the behavior of data (Rezk et al., 2017). This study describes a review of state-of-the-art literature on breast imaging, tumor detection, segmentation, classification techniques, and breast dataset. The papers published in the journal containing keywords of breast cancer detection, segmentation, classification, and breast dataset are extracted from different sources including Research Gate, Google Scholar, Pubmed, Elsevier, and Springer. The most common imaging modalities consisting of mammography, ultrasound, MRI, thermography, CT, and biopsy are used as keyword to find the relevant papers. Based on the titles and abstracts, about 80 between 2016 and 2020 papers are selected and studied.

The different sections of this study are as follow. The research methodology in Section 2 consisting of research questions and research framework. The standard breast data sets are presented in Section 3 and the paper is finally concluded in Section 4.

E-mail address: zrezaei@miau.ac.ir.

¹ ORCID: 0000-0001-5039-1474.

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Nomenclature			
ABUS	Automated 3D Breast Ultrasound	IBT	Infrared Breast Thermography
ANN	Artificial Neural Network	IoU	Intersection Over Union
AUC	Area Under The ROC Curve	IR	Infrared
BCDR	Breast Cancer Digital Repository	JSI	Jaccard Similarity Index
BCDR-DMR	Full Field Digital Mammography Repository	k-NN	K-Nearest Neighbor
BCDR-FMR	Film Mammography Repository	LBP	Local Binary Pattern
BUS	Breast Ultrasound	LDA	Linear Discriminant Analysis
CAD	Computer-Aided Diagnosis	MCC	Mathew Correlation Coefficient
CC	Cranial Caudal	MCs	
CFN	Cascade Forward Net	ME	Mean Absolute Error
CT	Computed Tomography	MG	X-Ray Mammography
DDSM		MIAS	
DEMS		MLO	
Dice	Dice Similarity Coefficient	MLP	Multilayer Perceptron
DT		MRI	
ELM	Extreme Learning Machine	NB	
FAR	False Alarm Rate	NPR	
FFN	Feed Forward Net	PCA	
FNAC	Fine Needle Aspiration Cytology	PN	Pattern Net
FNR	False Negative Ratio	PNN	Probabilistic Neural Network
FPF	False Positive Fraction	PPR	
FPR	False Positive Ratio	QDA	Quadratic Discriminant Analysis
FSC	Adaboost And Fuzzy Sugeno	RF	Random Forest
GFS	Genetic Fuzzy System	ROI	Region Of Interest
GLCH	Gray Level Co Occurrence Histogram	SI	Similarity Index
GLCM	Gray-Level Co-Occurrence Matrices	SIR	Similarity Ratio
GLRLM	Gray-Level Run-Length Matrices	SVM	Support Vector Machines
GM	Gm: Geometric Mean	TCGA	Cancer Genome Atlas
GT	Ground Truth	TPF	True Positive Fraction
HE	Hausdorff Error	TPR	True Positive Ratio
		UCSB-BB	UCSB Bio Segmentation Benchmark

2. Research methodology

This paper reviews the breast cancer diagnosis techniques for exploring the existing research gap. The research approach is influenced by the keywords, research scope, and the research framework. The scope is limited to the related journals from 2016 to 2020. Related studies of breast imaging techniques, tumor segmentation, feature extraction, and classification are investigated. Research methodology consisting of the research questions, and research framework.

2.1. Research questions

The following research questions (RQs) are presented to investigate the main objectives:

- RQ1: Which imaging techniques are used for tumor detection?
- RQ2: What are the essential factors for breast cancer detection?
- RQ3: Which breast segmentation techniques are frequently investigated by the researchers?
- RQ4: Which types of tumor features should be measured?
- RQ5: What are the present techniques for tumor classification?
- RQ6: What are the existing software for tumor detection?

2.2. Research framework

This section try to investigate the research questions by describing imaging modalities, the essential factors for breast cancer, various techniques of breast segmentation, feature extraction, tumor classification and existing software for breast cancer.

2.2.1. Which imaging techniques are used for tumor detection?

To detect the breast cancer, various breast screening methods have been developed consisting of the Breast Ultrasound (BU), mammography, Magnetic Resonance Imaging (MRI), Computed Tomography (CT), thermography, and biopsy, etc. (Saad et al., 2016). There are other imaging technologies that were rarely used due to challenge of complexity in diagnosis of breast cancer (Iranmakani et al., 2020).

2.2.1.1. Mammography. Mammography has been developed based on the X-rays imaging to generate the high-resolution images. This technique can diagnosis the abnormalities in overlapping tissue (Shi et al., 2018). The major task in digital mammography is to discriminate the malignant from the benign lesions (Xie, Li, & Ma, 2016). As the cancer region is masked by the normal tissue, this type of image is limited to have low sensitivity for patients with dense breast (Kozegar, Soryani, Behnam, Salamati, & Tan, 2017). Among the available diagnostic technique, mammography image is the most consistent technology to detect the benign and malignant masses in very early stage of cancer prognoses (Saad et al., 2016). The most common test to find the cancer, which is recommended for women aged 40 or above is screening by mammography (Berbar, 2018). Mammography produces three dimensional breast images consist of cranial caudal (CC) and the medial lateral oblique (MLO) views (Boone et al., 2006).

2.2.1.2. Breast ultrasound. Ultrasound or sonography image is explored as a dynamic, harmless, inexpensive, and commonly available technique for breast cancer detection. This modality used the reflected sound waves from the surface of tissue (Amin, Shahin, & Guo, 2016; Kandlikar et al., 2017). To define the tissue borders and understand the reflected signals, a transducer is used. In order to further examine the

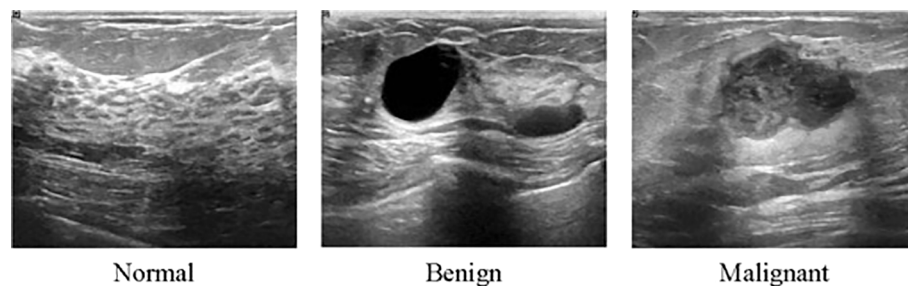


Fig. 1. Samples of Ultrasound breast images.

suspicious tissue during the mammogram or for discriminate between cysts and solid masses, ultrasound is commonly used. Furthermore, for supplementary screening the patients with dense breasts, automated 3D breast ultrasound (ABUS) can be employed (Kandlikar et al., 2017).

2.2.1.3. Magnetic Resonance imaging (MRI). MRI technique provides high resolution cross sectional images of breast using a strong magnetic field along with the pulsing radio. Screening breast by MRI technique is considered as a time consuming and expensive tasks, therefore it can be applied for screening the high risk cases or to better diagnosis of tumors (Kandlikar et al., 2017). The MRI data are used to detailed analysis of suspected tissues (Banaie, Soltanian-Zadeh, Saligheh-Rad, & Gity, 2018).

2.2.1.4. Computed Tomography (CT). The breast CT system scanning the breast tissue to produce the thick-section and high resolution images. The CT scan parameters are designated to use the equal radiation dose levels as mammography images. This methodology produces the structural detail of breast tissue, MCs representation, and visualization of the soft tissue of tumor (Boone et al., 2006). The problems of mammography techniques are resolved by means of removing the overlying and underlying anatomical tissue (Boone et al., 2006).

2.2.1.5. Biopsy. The histology or anatomical study of tissues, includes modification, processing, embedding, segmenting, and staining. To clearer visualization under a microscope, the tissue is stained by hematoxylin and eosin staining. The pathologist examines a biopsy section to identify the varied breast cancer, tumor growth, and so on (Aswathy & Jagannath, 2017). When the suspicious region is detected in mammography exam, a small sample of this region is taken by a fine needle aspiration biopsy (FNA) to allow the pathologist to investigate the type of cancer (Krawczyk, Galar, Jeleń, & Herrera, 2016).

2.2.1.6. Thermography. Thermography technique can detect the tumor faster than the conventional mammography imaging technology (Rastghalam & Pourghassem, 2016). Due to the using advanced infrared (IR) cameras and applying the computational tools for investigating the heat of breast, precision of thermography have considerably improved (Kandlikar et al., 2017).

2.2.1.7. Challenge of breast imaging modalities. Although, the ultrasound, mammography and breast MRI modalities are known as the most widely used breast techniques, but they have their limitations in breast cancer diagnosis (Xu & Li, 2020). The mammography tool has been demonstrated to be effective for early diagnosis of breast cancer and reduce the mortality. However, the examination to marks the abnormal region is performed manually by a radiologist can lead to medical-misdiagnosis (Ahmed et al., 2020). Moreover, in terms of the overlap tissue in mammography image, it can be used for examination the low-density breasts (Xu & Li, 2020). Most importantly, exposing to the X-ray radiation system increases the risk of breast cancer in women younger than 30 years old. Due to pressure during the imaging procedure,

mammography causes pain for sensitive breasts as well.

Because of the dense tissues, small tumors may be missed in mammograms, therefore handheld ultrasound technique can be used as a recommended method. However, this breast examination do not reliably identify the Microcalcifications (MCs), and the detection rate with this system depends on the operator skills. The error rate of cancer detection may be increased due to the simultaneously of collecting and interpreting the images (Kozegar et al., 2017). On the other hand, the quality of ultrasound image may be affected by the shadowing, speckle noise, echo, etc., due to limitation of imaging system (Flores & de Albuquerque Pereira, 2017).

Because of adverse effects of mammography on human body, biopsy and MRI images are developed (Aswathy & Jagannath, 2017). Although breast MRI provides clear images and high resolution of soft tissue signals, however it is limited to illustrate the calcification.

The breast CT demonstrates the detail of breast anatomy and assessments the exact anatomical location of lesions. The scope of CT analysis is superior to the mammography image, therefore, this tool is more sensitive to cysts, calcification, hemorrhage, and fat high. However, in comparison with other techniques, the CT scan is expensive, has radiation exposure, and may cause the risk of allergic reaction (Xu & Li, 2020). Due to the delivering doses of 30–40 mammograms, the CT technique is not very convenient for early diagnosis the low-risk patient (Iranmakani et al., 2020). Currently, existing screening tools cannot fully predict breast cancer unless biopsy technique (Pavithra, Ravichandran, Sekar, & Manikandan, 2018). Although the biopsy evaluation process plays a key role in identifying the early tissue biopsy clues, it requires a great deal of time and effort for skilled pathologists and also highly dependent on the pathologist's knowledge and consistency of the pathological reports. The various protocols used in the cancer diagnosis laboratory, the skills of pathologists in scanning the breast images, and the various staining methods are also challenging (Vo, Nguyen, & Lee, 2019). Thermal imaging is non-contact, non-invasive, and safe modality that provides the functional properties of tissue, but low contrast of normal tissues causes difficulties for discriminating the malignant masses from the benign ones (Xie et al., 2016). This method can detect early-stage cancer, but due to need a high expertise for manual analysis and low specificity, it can be used only as an assistant modality (Kakileti, Manjunath, Madhu, & Ramprakash, 2017). In addition, this technique is limited due to false-positive and false-negative rates (Pavithra et al., 2018).

2.2.2. What are the essential factors for breast cancer detection?

Breast tissues can be depicted as clusters of different intensities consisting of the fibro-glandular, fatty, and the pectoral muscle. However the abnormal area involves the calcification or ill-defined masses (Shi et al., 2018). Generally, the malignant refers to cancer cells and benign indicating the non-cancerous tissue are two major types of breast tumors (Alzubaidi et al., 2020). The malignant cancer is represented by clustered distributions of MCs with linear branching pattern involving more than 3 of MCs together that have the diameter less than 0.5 mm, while benign cancer is considered as individual MCs. MCs can be considered as the fundamental symptoms of breast cancer detection in

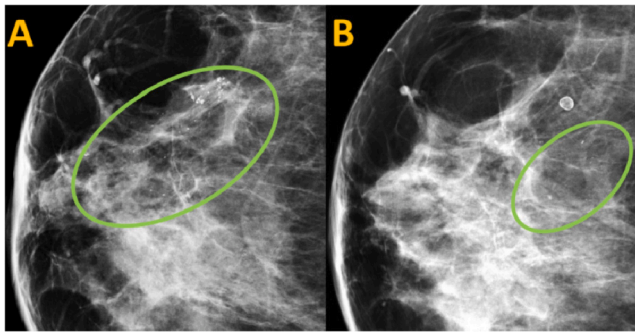


Fig. 2. (A) Previous mammogram of the right breast, (B) Current mammogram of the right breast.

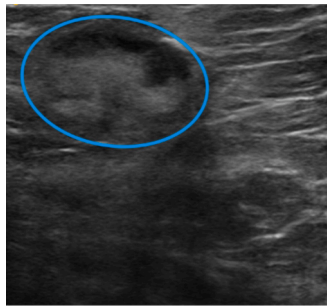


Fig. 3. The Ultrasound image demonstrates a morphologically abnormal lymph node.

mammograms. As a result, analysis and interpretation of MCs is important to early detection and decrease the mortality rates (Kandlikar et al., 2017). However, the issue of false negative diagnosis due to poor quality of the MCs causes missing elusive abnormalities (Saad et al., 2016). In order to categorize the MCs, important features consisting of size, shape, the density, location of the margins, number of MCs, and MCs distribution are considered. In benign tissue, MCs have the diameter of 1–4 mm with the round or ovule shape, and it is represented by the coarse or uniform texture of scatter pattern (Saad et al., 2016). The presence of masses and calcifications can be considered as the signs of cancer (Berbar, 2018). In order to early detection of breast cancer, the specialists should examined all tumors even they depict the normal tumor (Abd-Elrazek, Othman, Abd Elaziz, & Abd-Elwhab, 2018). Fig. 1 shows the normal, benign and malignant tissue of the original ultrasound images (Al-Dhabyani, Gomaa, Khaled, & Fahmy, 2020).

Malignant and benign masses include different characteristics that might be utilized for lesion classification (Xie et al., 2016). A breast lump or mass is commonly considered as the sign of breast cancer. The tissue of breast cancer can be harder than the normal and benign tissue (Zhang et al., 2016). A painless mass with irregular shape are probably the most symptoms of cancer, however it can be tender, soft, round and even be painful. Fig. 2 depicts the current and previous mammograms of the right breast. Based on these images an unusual finding calcifications is shown in current image (Nguyen, Nguyen, Dixon III, Monetto, & Robinson, 2020).

To investigate the uncommon behavior of calcifications, auxiliary ultrasound imaging was suggested and morphologically abnormal lymph node with eccentric cortical thickening was confirmed (see Fig. 3).

Moreover, tumor shape including irregular, round, lobular, and oval, also margin type consisting of circumscribed, ill defined, speculated, and obscured play a vital and important role in diagnosis of malignant tumors (Singh et al., 2020). However, the precise characterization of the shape is a challenging task (da Rocha, Junior, Silva, de Paiva, & Gattass, 2016). In order to assess the degree of cancer malignancy growth and

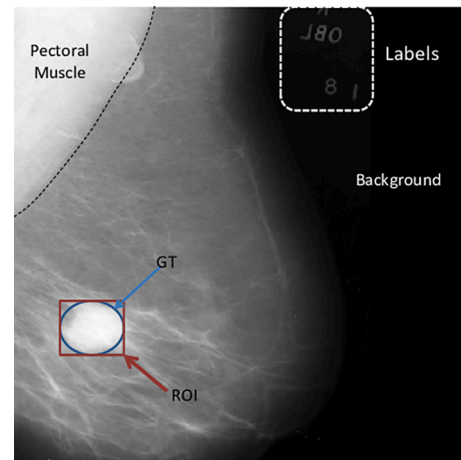


Fig. 4. ROI generation using GT (Abdel-Nasser, Rashwan, Puig, & Moreno, 2015).

elaborate a personalized therapy, level estimation is essential as well (Krawczyk et al., 2016).

2.2.3. Which algorithms are practical for breast segmentation?

The segmentation process should be considered as a serious step to identify the benign and malignant tumors (Rouhi & Jafari, 2016).

2.2.3.1. Preprocessing. The preprocessing step can be performed on breast image for elimination the unwanted regions consisting of margins, labels, and pectoral muscle (Raghavendra et al., 2016). Fig. 4 illustrates the pectoral muscle, labels, and tumorous ROI using the ground truth (GT) for mini Mammographic Image Analysis Society (mini-MIAS) database of mammographic.

The muscle region has similar intensity compared to the tumor, therefore it should be removed from the original image (Vijayarajeswari, Parthasarathy, Vivekanandan, & Basha, 2019). Fig. 5 illustrates the removed label and pectoral muscle in mammography (Raghavendra et al., 2016).

2.2.3.2. Extraction of the region of interest. In medical application, quantification of the suspicious area as region of interest (ROI) and classification of the positive and negative regions in real-time images are performed manually (Mohammed et al., 2018). Usually, a coarse outline and a lesion diameter size of ROI are supplied by the radiologist to illustrate the size and location of tumor (Lang, Sklair-Levy, & Spitzer, 2016). Boundary segmentation and calcification localization of the breast are necessary stages (Shi et al., 2018). The lesion segmentation can help to restrict the analysis of tumor regions and avoid the misperception with other areas (Rezk et al., 2017). Medical image segmentation can be a complicated procedure because of low quality and imaging artifacts (Rezaei, Selamat, Taki, Rahim, & Kadir, 2019). Automatic segmentation tools delineate the cancer borders and also it is useful to avoid the missing in mass detection (Lang et al., 2016). In order to classify the tumors, quantitative features including size, shape, echo pattern, etc. are extracted from the ROI and fed to the classifier. Therefore, the performance of tumor detection is directly affected by the accuracy of segmentation (Xian, Zhang, Cheng, Xu, Zhang, & et al., 2018). To detect the changes in nuclear surface, pathologists examine several biopsy images under a microscope daily. To address the restrictions of human eye in distinguishing between color and intensity image, segmentation algorithms are applied (Puri et al., 2016). Many studies have been investigated the challenge of breast segmentation by using the computational methods, including thresholding, morphology-based, zero-crossing, watersheds, clustering, classifier-based methods, Markov random field, active contours, level-set, texture-based methods,

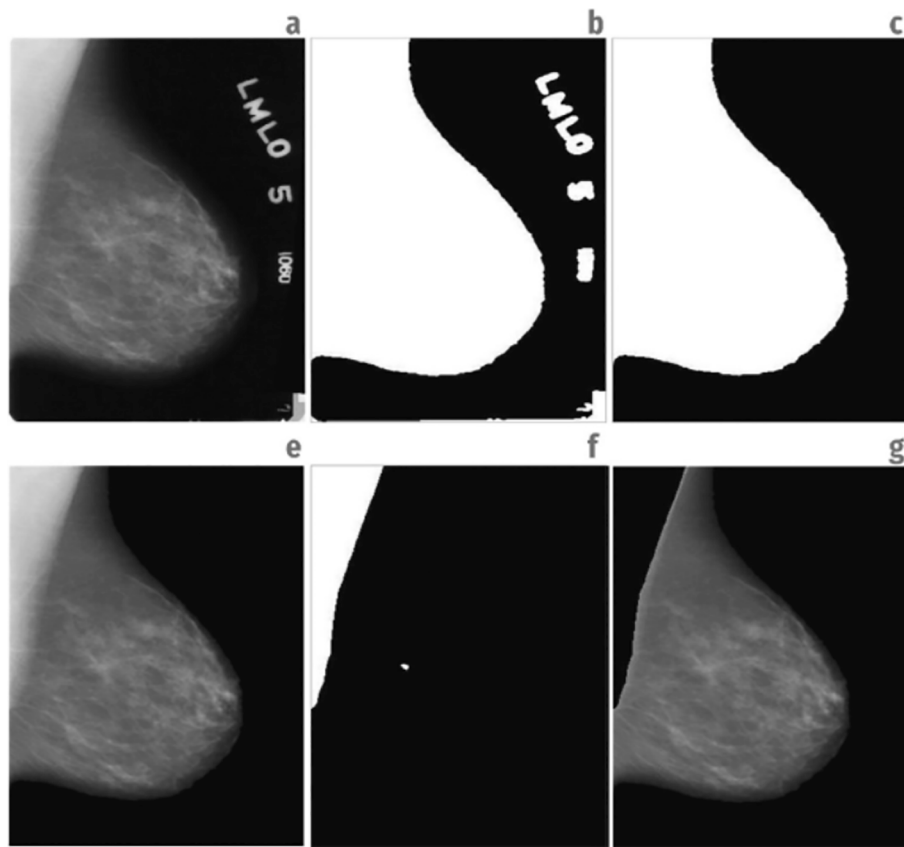


Fig. 5. Preprocessing steps in a mammography image, (a) original image, (b) binary image, (c) masked image, (d) removed label, (e) pectoral muscle, and (f) remove pectoral muscle.

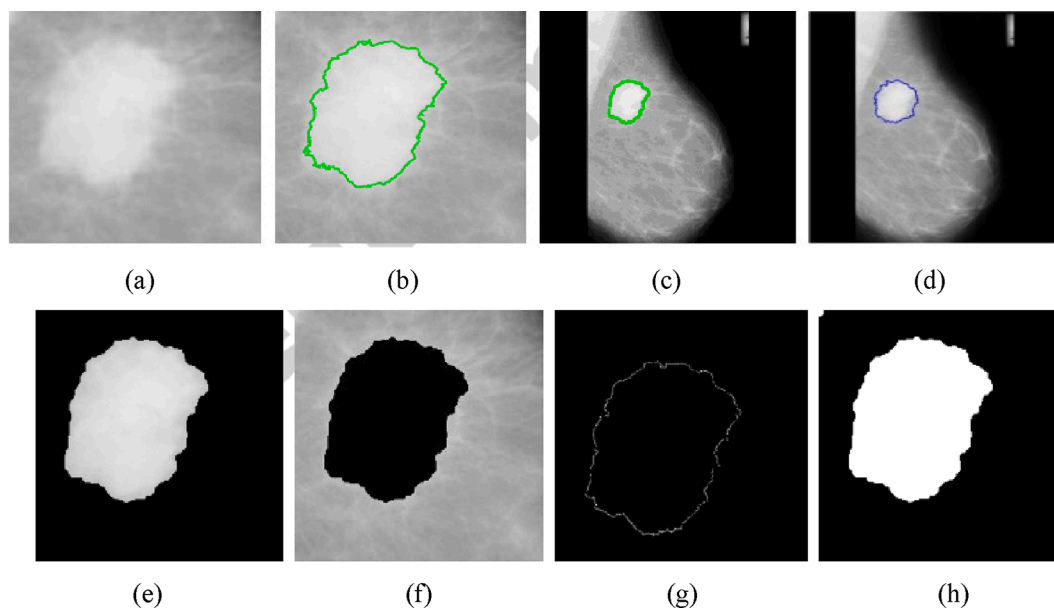


Fig. 6. Segmentation result: (a) the ROI of mdb184 image in MIAS database, (b) segmentation by Level Set, (c) mass segmentation in the original image, (d) segmentation result by the proposed model by Li, Xu, Gui, and Fox (2010), (e) the mass region, (f) ROI without the mass region, (g) the mass boundary, (h) the result of binarization (Xie et al., 2016).

and combining multiple different algorithms (Lang et al., 2016; Shi et al., 2018; Yang, 2017). An example of mass segmentation result using the Level Set method is presented in Fig. 6. Morphological operation is applied after segmentation for feature extraction (Xie et al., 2016). Table 1 illustrates the overview of breast lesion segmentation.

2.2.3.3. Challenge of breast image segmentation. Breast cancer image segmentation is complicated and difficult. Although different techniques have been proposed for this purpose, however, every study has limitations. The thresholding technique may causes the improper segmentation due to the overlapping area between subject and background

Table 1
Overview of breast lesion segmentation.

Authors	Methods of Segmentation	Image	Dataset	Validation	Quality Performance
(Keatmanee, Chaumrattanakul, Kotani, & Makhanov, 2017)	Active contour model (Snake)	Ultrasound	90 image collected by Thammasat University Hospital of Thailand.	The ground truth was hand-drawn by three leading radiologists of the hospital	Acc: 88.9%
(Xi et al., 2017)	Prior knowledge learning by SVM	Ultrasound	The 135 benign and 51 malignant cases from Qianfushan Hospital of Shandong Province.	The lesions were manually delineated by radiologists	Acc: 94.02% JSI: 61.7% TP: 67.81% FP: 17.1% JSI:0.81%
(Flores & de Albuquerque Pereira, 2017)	Adaptive contrast enhancement	Ultrasound	The 500 breast images of patients with posterior indication of biopsy collected from National Cancer Institute (INCa) of Rio de Janeiro, Brazil.	A senior radiologist manually delineated all breast lesions in the data set to approximate actual tumor shapes.	
(Osman & Yap, 2018)	Thresholding	Ultrasound	The 306 images, B&K Medical Panther 2002 and B&K Medical Hawk 2102, UDIAT Diagnostic Centre, Sabadell (Spain).	Expert radiologist processed each image manually, and marked the extreme points of the suspected lesions with crosses.	JSI: 75.09% Dice score: 86.85% MCC score: 86.62%
(Daoud, Atallah, Awwad, Al-Najjar, & Alazrai, 2019)	Graph cuts Active contour model	Ultrasound	The 86 benign and 74 malignant from Jordan University Hospital, Amman, Jordan	Manual tumor outline drawn by an experienced radiologist	TPR: 96.04%, FPR: 7.99%, FNR: 3.96%, SIR: 91.41%, HE: 11.66%, ME: 3.65% Dice:
(Rampun, Morrow, Scotney, & Winder, 2017)	Thresholding and Active Contour Models	Mammograms	• MIAS (322) • INBreast (208) • BCDR (100)	Three different databases including MIAS, INBreast, and BCDR were tested. Also evaluation based on ground truth was performed	MIAS :98.8% INBreast :98.9% BCDR: 99.2%
(Maitra & Bandyopadhyay, 2017)	Edge-based and region-based	Mammograms	• MIAS (322) • DEMS (485)	The ROC analysis using two public database and considering MIAS database as benchmark with ground truth images by the radiologist.	Acc: 96% Sen: 97.6% Sp: 88.6%
(Shi et al., 2018)	Gradient weight map	Mammograms	• MIAS (322) • BCDR (100) • INbreast	Quantitative and qualitative evaluations on public dataset	MIAS: Acc: 97.08%, JSI: 94.89% Dice: 96.40% BCDR: Acc: 97.91% JSI: 96.22% Dice: 97.66% INbreast: Acc: 97.38% JSI: 95.96% Dice: 97.60% Dice: 94% IoU: 87%,
(Singh et al., 2020)	Conditional Network	Mammograms	• DDSM • INbreast	Evaluation using two public (INbreast, DDSM) mammography datasets and one private dataset	
(Yang, 2017)	Support-pixel correlation statistical method	Computer-simulated images, Mammogram, and MRI	• Representative single-spectrum images (4) • MIAS (322) • MRI collected from Women's Health Center of Tri-service General Hospital, Taipei, Taiwan	Quantitative assessment using computer image simulation	Computer-simulated: FAR: 18 0.86% Acc: 84 0.25% Sp: 81 0.14% Mammograms : FAR: 7 0.87% Acc: 88 0.14% Sp: 92 0.12% MRI: FAR: 7 0.77% Acc: 91 0.26% Sp: 86 0.22% Se: 95.43% GM: 95.73% AUC: 96.38% JSI: 56.67% Dice: 71.66% Tanimoto: 56.67% Acc: 98.83%
(Krawczyk et al., 2016)	FCM, level set, and greylevel	Biopsy	Database of FNA slides includes 341 images collected at the Medical University of Wroc law, Poland.	Experiments and evaluation carried on a large biopsy dataset collected by the authors	
(Kumar, Srivastava, & Srivastava, 2017)	FCM	Microscopic biopsy	31 benign and 27 malignant images from a Benchmark Breast Cancer cell dataset	Evaluation by the region of interest selected ground truth available for Benchmark data set	

(Yang, 2017). The morphology-based approaches consider the shape model for fitting the objects of breast. However, using the natural shape features cannot cover the complex shapes of mammography images (Shi et al., 2018). Active Counter methods use the minimizing energy functions to determine the actual breast boundary (Shi et al., 2018). The

accuracy of this model can be affected by initialization stage (Yang, 2017). The classifier-based method need to use the appropriate features, also due to the providing the individual blocks, image resolution may be reduced (Yang, 2017).

Due to speckle noise, shadowing, and low quality in ultrasound,

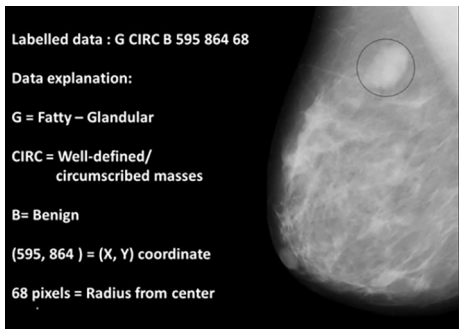


Fig. 7. Mammogram image with labelled data.

segmentation and classification is challenging (Osman & Yap, 2018). Moreover, because of the artifacts, homogeneity of pectoral and breast tissue, etc. estimating the boundaries of breast is difficult (Rampun et al., 2017). Furthermore, due to the overlapped tissue in ultrasound images, discriminate of the solid masses is also uncomfortable for the radiologists (Osman & Yap, 2018).

Some of the segmentation methods have performed the tumor detection based on the ground truth information. For example, the MIAS dataset includes the information of background tissue, the type of abnormality and its severity, x-axis, y-axis coordinates of center, and radius the abnormality. Fig. 7 illustrates a labelled ground truth as a circles on a raw digital mammogram image using labeled data (Ting, Tan, & Sim, 2019). The restricted area can be used as initial points by Active Counter methods. However, these coordinates are not available for all dataset. Therefore, selection of initial points and ROI detection should be performed automatically. Moreover, in some breast images, the ROIs may include part of pectoral muscle as these two regions are close to each other (Abdel-Nasser et al., 2015).

2.2.4. Which types of tumor features should be measured?

2.2.4.1. Feature extraction. Feature extraction used for reducing the dimensionality of large image sizes and efficiently representing the

compressed vector. Assessment of the breast tissue structures required capturing the appropriate features (Krithiga & Geetha, 2020). Since the contour is not always precise in some lesions, to classify the benign and malignant tissues and more precise diagnosis, the textural features can be supplied from the ROIs without need of precise segmentation (Muramatsu, Hara, Endo, & Fujita, 2016). These approaches apply the texture filters consisting of wavelet, Gabor filter, gray-level Co-occurrence matrices (GLCM), gray-level run-length matrices (GLRLMs), Local Binary Pattern (LBP), etc. to determine the boundaries of the ROI (Shi et al., 2018). The studies reported that texture features provide a good description of patterns in breast images (de Oliveira, de Carvalho Filho, Silva, de Paiva, & Gattass, 2015). Fig. 8 depicts the Gabor wavelet features from the mammography image (Raghavendra et al., 2016).

2.2.4.2. Feature selection. Due to high dimensional feature space, the feature selection algorithm could be used to reduce the size of features, eliminate the irrelevant information, and reduce the entire training time (Krithiga & Geetha, 2020). Generally, several feature selection methods are introduced including principal component analysis (PCA), linear discriminate analysis (LDA), forward selection (FF), Fischer method, bidirectional search algorithm, and Convolutional Neural Network, etc. (Amin et al., 2016).

2.2.4.3. Challenge of feature extraction and feature selection. The previous schemes of computerized methods depend on the shape and margin features, however these features are provided from the ROIs without need of precise segmentation of breast lesion (Muramatsu et al., 2016). To describe the breast images and avoid the explicit segmentation, texture representation can be performed. Biomedical texture operations analyse the desired texture information in terms of spatial scales and the directions of a texture image, which are fundamental for visual texture discrimination (Rezaei et al., 2018). However, some type of breast images present the intrinsic heterogeneity and complex tissue structure (Spanhol, 2018b).

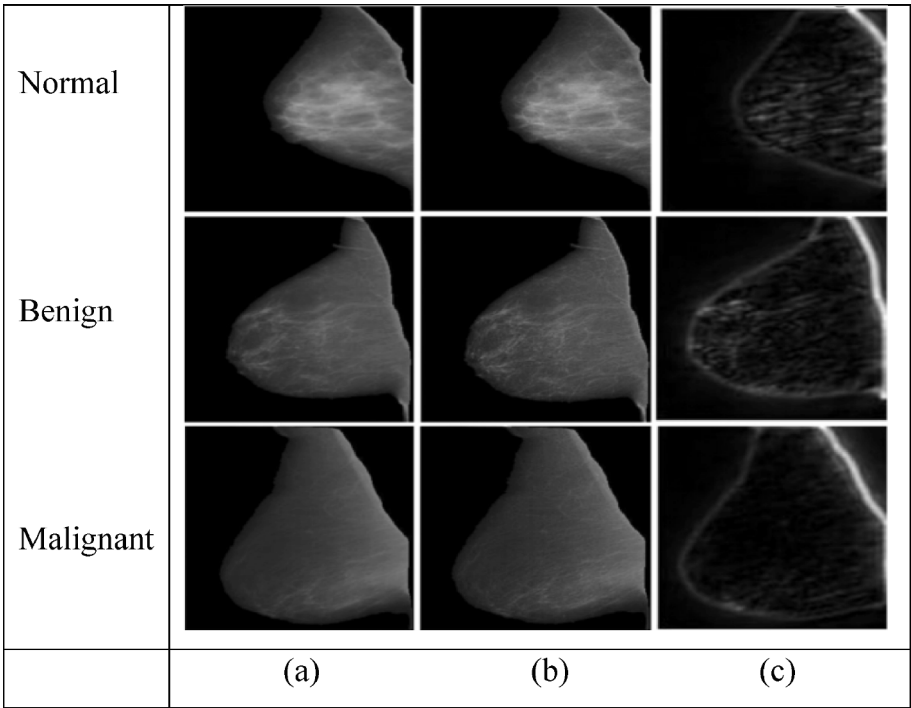


Fig. 8. The Gabor wavelet features from the mammography image, (a) preprocessed image, (b) enhanced image, (c) Gabor wavelet transformed image.

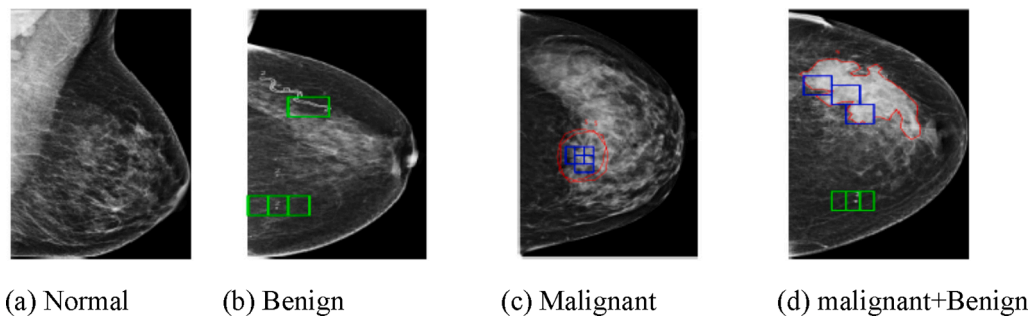


Fig. 9. Classification on the breast.

Table 2

The overview of tumor classification techniques for ultrasound images (Acc: accuracy, Sn: sensitivity, Sp: specificity, Pr: precision, and FS: F-score).

Author	Objective	Features	Feature Selection	ClassificationMethod	Dataset	Validation	Performance
(Singh et al., 2016)	To categorize benign and malignant tumors	Textures and shapes	Multi-criterion	Combination of FCM and BPANN	The 88 benign and 90 malignant cases from two hospitals in India and online data	Comparison with SVM and traditional BPANN	Acc: 95.862% Sn:95.13% Sp:96.57% AUC:95.85% MCC:9.73%
(Gómez-Flores & Ruiz-Ortega, 2016)	To distinguish between the normal tissue and breast lesion	Correlation Covariance coefficients, GLCM and fractal features	Averaging and quantization	LDA	544 BUS images from INCa of Rio de Janeiro, Brazil.	Lesion was outlined by a radiologist using a software (Oracle, Redwood City, CA, USA)	Benign MCC: 85% AUC: 92% Sn: 87% Sp: 97% Malignant MCC: 85% AUC: 91% Sn: 85% Sp: 97%
(Amin et al., 2016)	To categorize benign and malignant tumors	GLCH and Morphologic	Chi_squared technique	SVM	112 images from of Harbin Medical University, China	MLP-NN classifier used for performance comparison	Acc: 99.1%
(Kozegar et al., 2017)	To detect cancers	Gabor, GLCM, LBP, and basic features	No	Random Under-Sampling Boosting (RUSBoost)	112 cancerous and 33 benign from Radboud University Nijmegen Medical Centre	Performing statistical comparison with bootstrapping	Sn: 90% Pr: 0.41% Sp: 98% F1: 51% MCC:52%
(Mohammed et al., 2018)	To classify the benign and malignant breast tumor	Multi-fractal dimension	No	BPNN	The 184 breast from Hospital in Baghdad, Iraq	Comparison with manual readings generated by the experts	Pr: 82.04% Sn: 79.39%, Sp: 84.75%
(Moon et al., 2018)	To diagnosis breast cancer	Morphological features and GLCM	Backward feature selection	linear logistic regression	249 images from 247 patients form , Seoul, Republic of Korea	Evaluation with various feature sets	Acc : 75.1% Sn: 79.0% Sp: 71.5% PPV:71.8% NPV:78.8% Az: 75.7%
(Virmani & Agarwal, 2019)	classification of breast tumors into benign and malignant	Texture and morphological features	PCA	SVM	Total 100 images from a standard benchmark database of ultrasound	Evaluation with various feature sets	Acc: 96.0%

2.2.5. Which techniques are applied for tumor classification?

2.2.5.1. Breast tumor classification. Advance in breast cancer research for early detection and prognosis of breast cancer is really necessary. In order to classify the breast tissue and supporting the medical decisions, machine learning approaches have been widely used (Krithiga & Geetha, 2020). Commonly, the classifiers can be trained and used to classify the objects using the feature set (Krithiga & Geetha, 2020). Classification of the micro calcification clusters in breast images plays an essential roles in CAD system (Ren, 2012). Machine learning methods have been widely developed for classification of suspicious region into the normal, benign or malignant (Sheikhpour, Sarram, & Sheikhpour, 2016). A combination of unsupervised and supervised learning was proposed by Singh, Verma, and Thoke (2016). The extracted features can be

classified by applying the Artificial Neural Networks (ANNs), Support Vector Machines (SVM), k- Nearest Neighborhood (k-NN), Naïve Bayes (NB), and random forest (RF), and Genetic Fuzzy System (GFS), and extreme learning machine (ELM) approaches (Rouhi & Jafari, 2016). In order to improve the lesions classification in CAD systems, ensemble classifier structure can be performed (Magna et al., 2016). Fig. 9 illustrates the classification of the breast tissue into normal, benign, and malignant (Bakalo, Goldberger, & Ben-Ari, 2019). Tables 2–4 depict the overview of tumor classification techniques. These tables are divided based on the different imaging modalities including BU, mammography, MRI, CT, thermography, and biopsy.

Noticeably, performance of classification methods depends on the useful feature and data representation (Spanhol, 2018a). In some of the proposed methods, although texture-based methods used to extract the breast feature, but the feature selection methods were not employed to

Table 3

The overview of tumor classification techniques for mammography images (Acc: accuracy, Sn: sensitivity, Sp: specificity, Pr: precision, and FS: F-score).

Author	Objective	Features	Feature Selection	ClassificationMethod	Dataset	Validation	Performance
(de Lima, da Silva-Filho, & dos Santos, 2016)	to detect and classify lesions to normal, benign, and malignant	Multi-resolution wavelets and Zernike moments	No	SVM, MLP, and ELM network	- IRMA - Mini-MIAS - DDSM	Compare with state-of-the-art classifiers: Decision Tree and K-NN	Acc: 94.11%.
(Raghavendra et al., 2016).	classify of the lesions to normal, benign, and malignant	Gabor Wavelet	Locality Sensitive Discriminant Analysis (LSDA)	DT, LDA, QDA, k-NN, NBC, PNN, SVM, FSC	DDSM	Evaluated using 690 mammogram images taken from a benchmarked dataset	Acc: 98.69%, Sn: 99.34% Sp: 8.26%
(Gedik, 2016).	Classification of normal or abnormal and then benign or malignant	Fast Finite Shearlet Transform (FFST)	Ranking <i>t</i> -test and thresholding	SVM	- MIAS - DDSM	Comparison with previous studies of shearlet transform	MIAS: Acc:97% DDSM : Acc: 100%
(da Rocha et al., 2016)	Classification of benign or malignant	LBP, image histograms, GLCM, GLRLM and diversity index	No	SVM	DDSM	Visual analysis by graphics for texture features	Acc:88.31% Sn: 85% Sp: 91.89% PPR: 10.48% NPR: 0.16% ROC: 0.88%
(Bruno et al., 2016)	Classify the normal-malignant and benign-malignant	Curvelet transform , LBP	Statistical analysis of variance (ANOVA)	DT, RF, SVM, and polynomial (PL)	- DDSM - BCDR-FMR, BCDR-DMR and UCSB-BB	Evaluation with four breast cancer datasets	AUC: 91% Acc: 100%
(Muramatsu et al., 2016)	Classification of benign and malignant masses	Radial local ternary patterns (RLTP), GLCM, DWT	No	ANN, SVM, and RF	376 digital images from Nagoya Medical Center, Nagoya, Japan	Evaluation with conventional rotation invariant patterns	AUC: 90%
(Saad et al., 2016)	To classify normal, benign or malignant tissues	GLCM Kurtosis Skewness	No	ANN	- DDSM - MIAS - Local database	Comparison with Adaboost	Acc: 98.68% Sn: 80.15%
(Rouhi & Jafari, 2016)	To recognize benign and malignant	Intensity, shape and texture	GA	ANN, RF, NB, SVM, and k-NN	MIAS	Comparison with existing methods for breast cancer diagnosis	Sn:100% Sp:97.30%
(Magna et al., 2016)	Tot diagnosing bilateral asymmetry	Cross- correlation analysis in both spatial and wavelet domains	PCA	Ensemble of Adaptive Artificial Immune Networks	- DDSM - mini-MIAS	Comparison with gold standard classifier	Acc: 90% Sn: 93% Sp: 87%
(Choi, Kim, Plataniotis, & Ro, 2016)	Classify masses and normal tissues classification between malignant and benign masses	Texture, intensity, shape, margin, speculation	Random feature selection	Ensemble classifier framework	DDSM	Compare with single SVM and the single NN	AUC: 93.2%
(Berbar, 2018)	Classify tissue into malignant or normal	GLCM, DWT	No	SVM	- DDSM - MIAS	Compare with most recent researcher's methods.	Acc: 98.31% Sn: 98.82% Sp: 98.45%
(Rampun, Scotney, Morrow, Wang, & Winder, 2018)	Breast density classification	Local Quinary Patterns (LQP)	select the dominant patterns by histogram	SVM	- MIAS - InBreast	Compare with some of the best accuracies reported in the literature.	Acc: 86.13% Sn: 80.67%, Sp: 92.81% AUC:89.92%
(Goudarzi & Maghooli, 2018)	Distinguish between normal and abnormal classes	Geometric and Texture	Discretization and conceptual hierarchy	Fuzzy	mini-MIAS	Combination of three classifiers	Acc: 89.37%
(Vijayarajeswari et al., 2019)	Classification of normal and abnormal tissue	Hough transform	No	SVM	MIAS	Compare with LDA	Acc: 94%

reduce the volume of features. Also, some of studies were not used the public data, so the expert should be available for the validation. According to the above tables, mammography, and ultrasound imaging modalities with public dataset are frequently used for tumor classification. The portion of each classifier is illustrated in Fig. 10. As shown in this figure, SVM and ANN are frequently used and HMM, Fuzzy, and PL rarely employed. Moreover, the desirable accuracy is obtained by applying the SVM or multi classifiers.

2.2.5.2. Challenge of breast tumor classification. It should be noted that,

due to the employing various methodologies, variations between databases of image, different number of sample images for tests, and types of evaluation metrics, the evaluation of results in not an easy task (da Rocha et al., 2016). One of the main challenges is lack of standard image dataset for some of imaging modalities. Therefore, in order to evaluate their methods, visual assessment by a radiologist must be performed (Alzubaidi et al., 2020). Comprehensive and public image databases are crucial to develop and validate the machine learning methods (Spanhol, 2018b). Moreover, lack of scalability of the clinical dataset, low training samples, imbalanced class distribution, and existences the outliers

Table 4

The overview of tumor classification techniques for MRI, thermography, and biopsy Images (Acc: accuracy, Sn: sensitivity, Sp: specificity, Pr: precision, and FS: F-score).

Image	Author	Objective	Features	Feature Selection	ClassificationMethod	Dataset	Validation	Performance
MRI	(Fan et al., 2017)	accurate prediction of the breast cancer	FO, GLCM, Dynamic, and morphologic features	Evolutionary algorithm (EA)	logistic regression classifier	57 cancer patients of DCE-MRI from Hospital of Chongqing Medical University	Quantitative analyses	AUC: 68.3% ROC: 70.3%
	(Lu, Li, & Chu, 2017)	Automated computer-aided diagnosis (CADx)	Morphological, texture, and Gabor features	Relief algorithm	Ensemble classification	438 positive and 1898 negative instances, Shandong Cancer Hospital in China	Test with a manually segmented ROI	Sn: 90.19% Sp: 96.31% Fs: 87.46% G-mean: 93.15% FPC: 3.69% AUC: 96.17% Acc: 99%
	(Banaie et al., 2018)	Classification of benign and malignant	GLCM, Shape, and temporal	Ranking by <i>t</i> -test	ANN	The DCE-MRI data of 26 biopsy or short-term follow-up from (GE Healthcare, Milwaukee, WI, USA)	Compare with SVM	
Thermography	(Rastghalam & Pourghassem, 2016)	To classify normal and cancer	New scheme of LBP and MRF	No	HMM	65 breast images available from the published papers	Evaluation on a variety dataset of thermography	PPV: 0.95% NPV: 0.95% Sn: 0.95% Sp: 0.92%
	(Gogoi, Majumdar, Bhowmik, & Ghosh, 2019)	To characterize healthy, benign and malignant breast	Intensity and temperature features	No	SVM	The 60 images (25 healthy, 23 benign, and 12 malignant cases), India	To evaluate, clinical examination, MG and FNAC reports were collected	Acc: 83.22% Sn: 85.56% Sp: 73.23%
Histology	(Rezk et al., 2017)	To classify benign and malignant tissue	Stratified sampling	No	NB, SVM, PN, CFN, and FFN	MITOS	Evaluation by a machine-learning case study of breast cancer images	Acc: 78% Fs: 0.84%
	(Sun, Li, Tang, & Wang, 2018)	To improve the prediction of breast cancer	Genomic data, image features	FSelector package	Multiple-kernel version of SVM	Cancer Genome Atlas (TCGA)	Compared with existing breast cancer survival prediction method	AUC: 82.8%

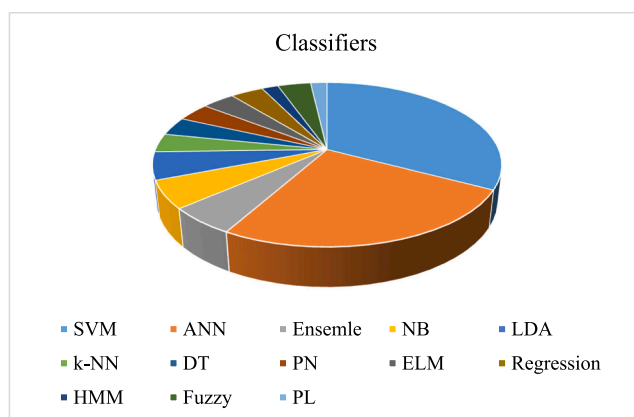


Fig. 10. The portion of each classifier for breast tumor classification.

would affect the outcome (Ahmed et al., 2020). In addition, find a decision boundary between two classes is quite challenging due to fact that appearance of breast and pectoral muscle is approximately similar. In addition, classifiers need to have a large amount of data (and ground truth) to build a reliable network. Unfortunately, obtaining ground truth data from a radiologist is difficult and time consuming (Rampun et al.,

2017).

2.2.6. What are the existing software of breast cancer detection?

In order to analyze the mammograms for cancer detection, CAD technology which is approved by FDA is developed in 1998 by applying the image processing techniques and artificial intelligence algorithms (Saad et al., 2016). The specialized radiologist determines the top, left, bottom, and right boundaries of a rectangle. Then, the ROI is presented to a CAD for segmentation and classification of breast tumor (Osman & Yap, 2018). In order to provide the clinical evidences for experts in this system, the breast features should be automatically measured (Shi et al., 2018). The tumor composition is classified using the Breast Imaging Reporting and Data System (BI-RADS) into category I (25%), category II (50% of), category III (50–75%), and category IV(75%) of glandular density (de Oliveira Silva, Barros, & Lopes, 2017). However due to breast tissue compression during the mammography image acquisition, the precise detection of mass shape in the image is a difficult task for image processing techniques (da Rocha et al., 2016). A free open source tool, namely CellProfiler (Carpenter et al., 2006) is applicable to pathological images and is developed to help the researchers in order to characterize the features in images.

3. Breast image dataset

In this section, some of the publicly available breast cancer dataset are described.

BUS image Segmentation:

A benchmark of B-mode BUS image segmentation consisting of 562 images were published by Xian, Zhang, Cheng, Xu, Huang, &, et al. (2018). Total 100 breast ultrasound images (40 benign and 60 malignant) taken from a standard benchmark database of ultrasound images available at <http://ultrasoundcases.info/category.aspx?cat=67> [accessed December 2016]. Similarly, the 780 grayscale US images in DICOM format were collected at Baheya hospital. The dataset includes normal, benign, and malignant classes (Al-Dhabyani et al., 2020). Data accessibility <https://scholar.cu.edu.eg/?q=afahmy/pages/dataset>.

Histopathological Dataset: Spanhol (2018a) introduced a publicly available histopathological dataset namely BreakHis contains the breast tumors.

Cancer Genome Atlas (TCGA): The Cancer Genome Atlas (portal: <https://portal.gdc.cancer.gov/>) consisting of the gene expression, CNA, gene methylation, protein expression, and pathological images (Tomczak, Czerwińska, & Wiznerowicz, 2015).

MIAS (Mammographic Image Analysis Society): Mini-MIAS digital mammogram database including 322 Portable Gray Map (PGM) images representing the 8 bit, digitized with 200 m resolution and resized to 1024 × 1024 pixels (Suckling et al., 2015). A record containing tissue type, anomaly class, anomaly type (benign or malignant) and its center coordinates was presented (de Oliveira Silva et al., 2017).

INbreast dataset: A publicly available database having 410 mammography images from 115 cases includes masses, calcifications, asymmetries, and distortions cases. However, only 106 mammograms have ground truth matches for evaluation the detection and segmentation model (Singh et al., 2020).

DEMS (Dokuz Eylul University Mammography Set): DEMS database contains 485 images in DR / 16 bit category (Maitra & Bandyopadhyay, 2017).

MITOS: This dataset contains 50 images of 512 × 512 pixels of breast cancer biopsy slides and manually labelled as malignant or benign by domain experts (Rezk et al., 2017; Roux et al., 2013).

DDSM (Digital Database for Screening Mammography): This dataset consisted of 2500 cases includes 4 images for each cases (Bowler et al., 1996). The resolution of digitized images are between 42 and 50 m kept in LOSSLESS JPEG compression, 16 bits. The details of breast density, type and location of the lesions are recorded (de Oliveira Silva et al., 2017).

4. Conclusions and future work

This study described the available breast imaging systems and state-of-the-art literature on breast cancer detection and classification techniques. The existing software for breast cancer diagnosis and public datasets are introduced as well. The main issues and gaps in the existing approaches are discussed. According to the reported studies, the main challenges in present techniques include complexity, cost, human-dependency, and inaccuracy. Although considerable efforts have been made to improve the breast imaging, but due to breast positioning issue and variances in instrument settings, the enhanced resolution may fail to detect the breast cancer at a very early stage. It is important to highlight that the current imaging methods might not be appropriate for accurate diagnosis, therefore developing a low-cost, non-radiative, harmless, and portable tool is desirable. On the other hand, a combination of two available modalities such as ultrasound and mammography for each patient can be practical. In addition, supplementary information of the patients, such as age, family history of breast cancer, etc., should be available with the related images to accurately diagnosis the breast cancer. However extracting information, analyzing, and applying machine learning for large image dataset are difficult and require big data

framework and computer clustering. Therefore, an appropriate feature extraction techniques and classification in big data analytics would be discussed as future directions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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